

InnoDB and MyISAM: How to Choose Between These Storage Engines

Storage engines are the underlying software components of MySQL databases. They handle the SQL operations on various tables in the database. InnoDB and MyISAM are two of the many storage engines powering various databases. Each has its strong points and its weaknesses. Discover more about them in this article.



MyISAM and InnoDB are two popular storage engines. In a database management system (DBMS), a storage engine is the key software used to manipulate data by using the CRUD (create, read, update and delete) operation. The storage engine used by the database is vital to its performance, and particularly, its tables. Various storage engines provide better performance in certain situations, compared to others. For normal use, there are two competitors to be considered—MyISAM or InnoDB.

MyISAM is the default MySQL storage engine (prior to the 5.5 version) and InnoDB is an alternative engine built into MySQL intended for high-performance databases. In order to understand the differences between the two storage engines, we first need to learn more about each one of them.

An overview of MyISAM

MyISAM is the default storage engine

for the MySQL relational database management system versions prior to 5.5, and was released in December 2009. It is based on the older ISAM code. Each MyISAM table is stored on the disk in three files (if it is not partitioned). The files have names that begin with the table name and have an extension to indicate the file type. MySQL uses a *.frm* file to store the definition of the table. However, this file is not a part of the MyISAM engine, but a part of the server. The data file has a *.MYD* (MYData) extension. The index file has a *.MYI* (MYIndex) extension.

The advantages

1. Simple to design and create, MyISAM is better for beginners. Users need not worry about the foreign relationships between tables.
2. Faster than InnoDB on the whole, as a result of the simpler structure, and hence results in much lower costs for server resources.
3. Full-text indexing.

4. Especially good for read-intensive (select) tables.

The disadvantages

1. No data integrity check (e.g., relationship constraints), which then becomes a responsibility and overhead of the database administrators and application developers.
2. Doesn't support transactions, which is essential in critical data applications such as that of banking.
3. Slower than InnoDB for tables that are frequently being inserted into or updated, because the entire table is locked for any insert or update.

An overview of InnoDB

Since the release of MySQL 5.5.5 in 2010, the InnoDB storage engine for the database management system became the default storage engine in MySQL. It comes with the standard ACID-compliant transaction features, along with foreign key support (declarative referential integrity). It is included as

